Semester Test 2

Electricity and Electronics EBN111

15 May 2024

Assessment Information			
Maximum marks:	46	Full marks:	45
Duration of exam:	90 min	Open/Closed book:	Closed
OCR time allocations:	5 min before and 10 min after for OCR answer sheet completion	Total number of pages (this page included):	21

Lecturers: Prof. X. Ye, Mr. J. Thiselton, Mr. L. Staphorst

Assistant Lecturers: S. Twala, L. Hurter, J. Npembe

Important

1. The examination regulations of the University of Pretoria apply.
2. The departmental rules relevant to electronically graded assessments apply.
3. Answer all questions on the OCR answer sheets provided. The question number **01** in this question paper refers to row 01 of OCR answer sheet 00/01.
4. Read the instructions on OCR answer sheet 00/01 carefully.
5. Where applicable, answers must include units.
6. Final answers must be written as real numbers and not as fractions.
7. Use **at least 4 significant digits** to write real numbers.
8. Exact numbers in which decimal expansions are zero, can be written without the zero expansions, for example 0 V instead of 0.000 V or 3.5 A instead of 3.500 A.

INSTRUCTIONS

Do step 0 in the first 5 minutes of the assessment. (-5 to 0 min)

Do step 1 in the next 90 minutes of the assessment. (0 to 90 min)

Do step 2 in the next 10 minutes of the assessment. (90-10 min)

STEP 0 (-5 to 0 min) Complete the front page of the OCR form, and also fill in your name and student number on additional pages of the OCR form.

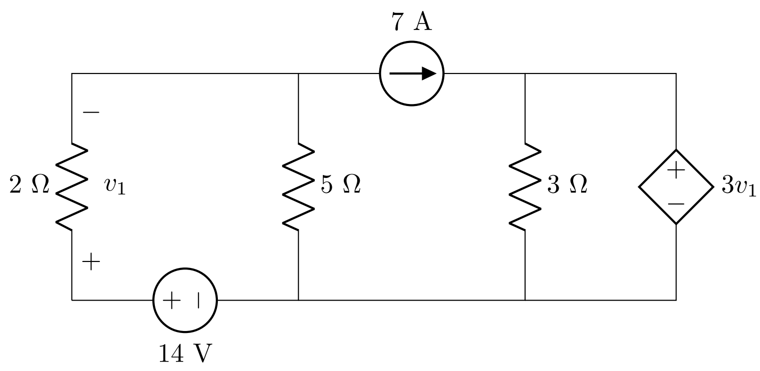
STEP 1 (0 to 90 min): Do your own calculations on the question paper. Write down the answers for numeric questions on the PRACTICE ANSWER SHEET. Be sure to include the units where applicable.

STEP 2 (90 to 100 min): Carefully copy your answers to the OCR answer sheets. Be sure to include units where applicable.

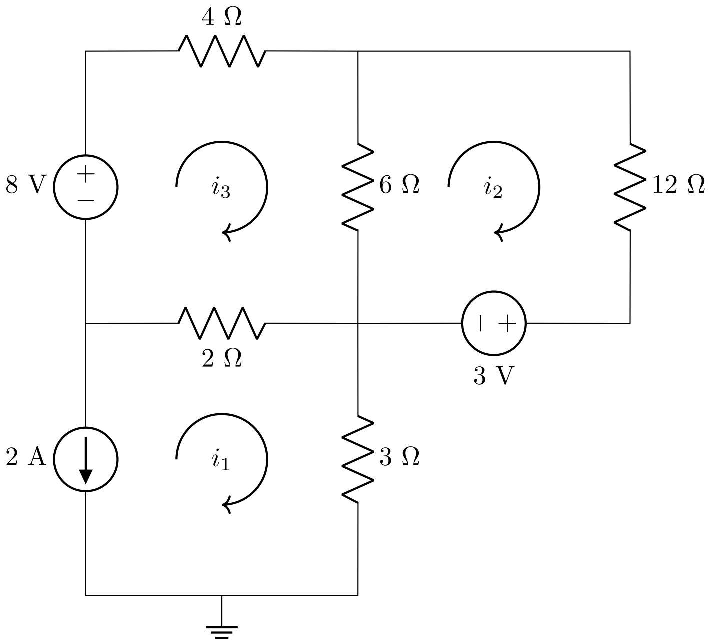
- The Assessment ID is: **2024EBN111S02**
- Use a mechanical pencil with 0.5 mm HB lead to complete the form.
- Each cell should only have one character, and characters may not touch or cross the cell.
- **Do not use commas!** Please make use of full stops as decimal points.

Section 1

Questions 1 to 2 [4]



01	Determine v_{1A}, the exclusive contribution of the independant current source to v_1. [2]
02	Determine v_{1B}, the exclusive contribution of the independant voltage source to v_1. [2]

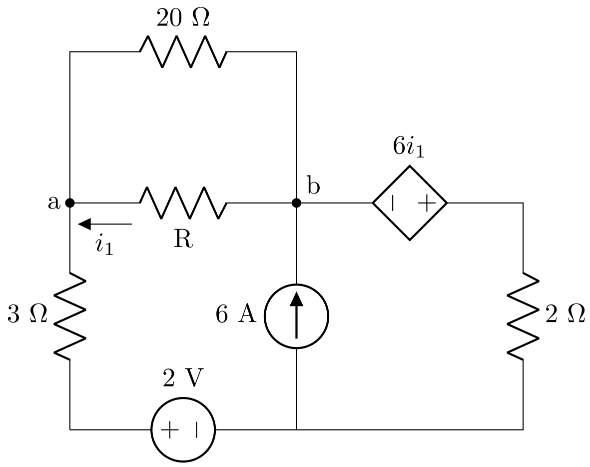
Question 3 [1]
Given $i_1/i_2 = -10$, find i_2.


03	i_2 [1]

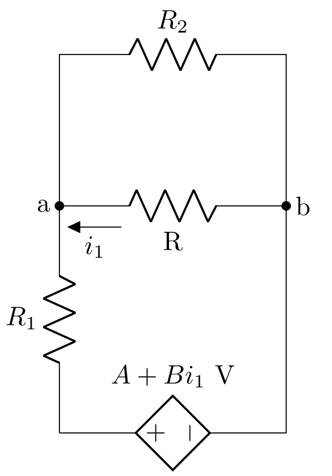
Questions 4 to 5 [3]

Simplify the following circuit A so that it is in the form of circuit B. Determine B and R_1.

Circuit A



Circuit B



04	B [2]
05	R_1 [1]

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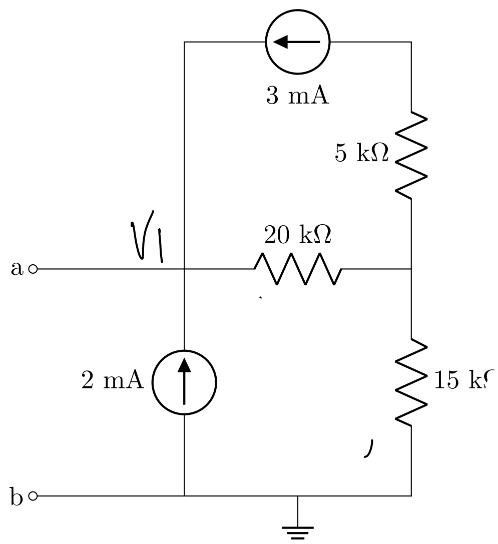
Questions 6 to 7 [3]

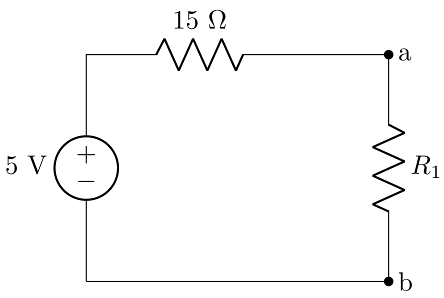
Calculate the Thevenin equivalent voltage V_{Th} and resistance R_{Th} with respect to terminals a-b.



06	V_{Th} [2]
07	R_{Th} [1]

Question 8 [1]

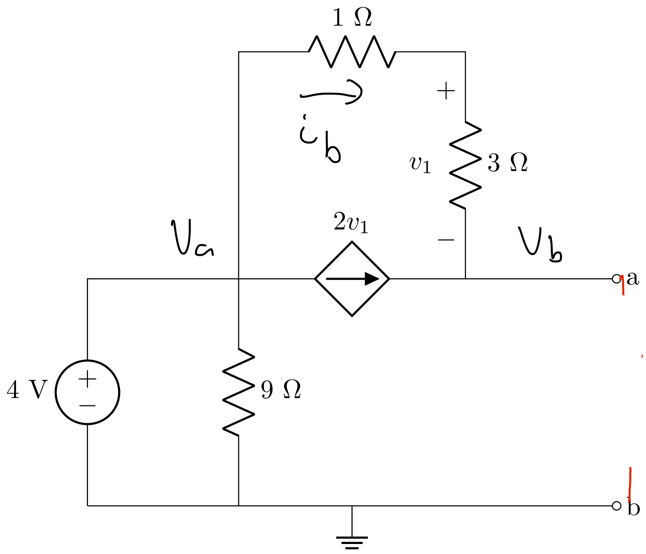
The following circuit contains a Thevinen equivalent (V_{Th} and R_{Th}) connected to a load R_1. What is the maximum power that can be transferred to R_1?



08	P_{max} [1]

Questions 9 to 10 [4]

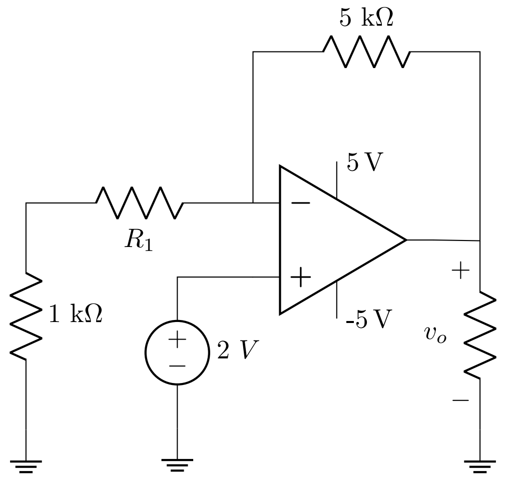
Calculate the Norton equivalent current I_N and resistance R_N with respect to terminals a-b.



09	I_N [2]
10	R_N [2]

Questions 11 to 12 [4]

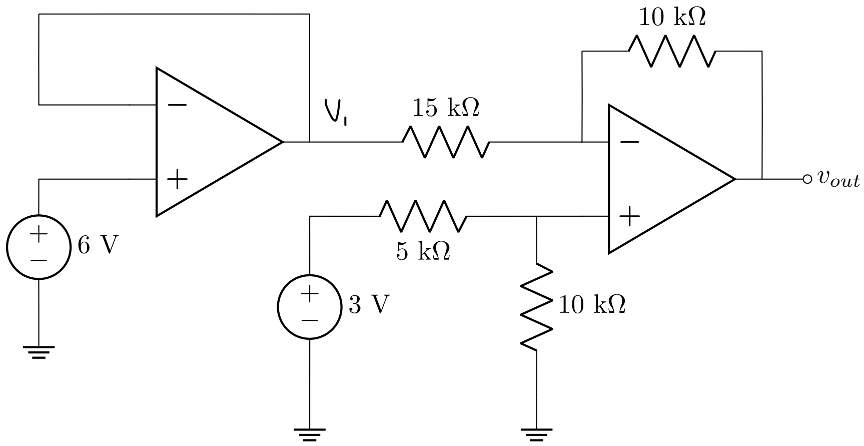
Assume the op-amp is ideal.



11	Calculate v_o when $R_1 = 3\ \text{k}\Omega$ [2]
12	Find the value for R_1 so that the op-amp is just entering positive saturation. [2]

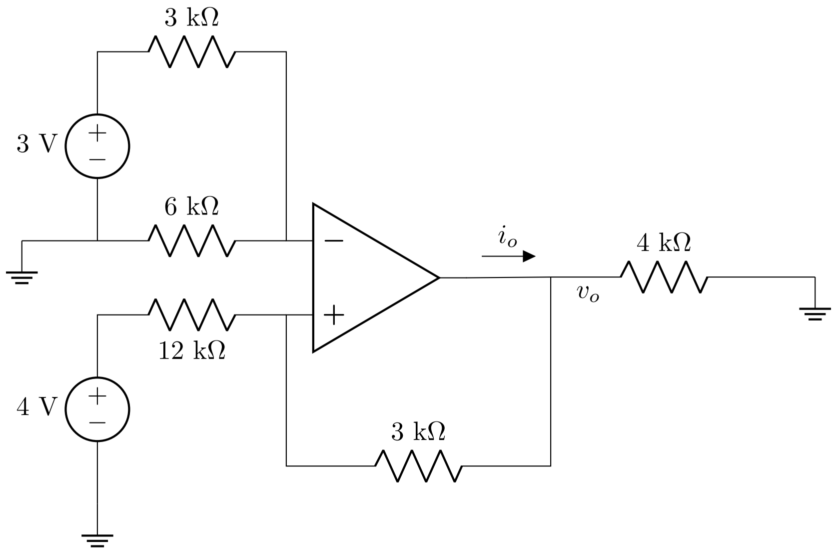
Question 13 [2]

Assuming the op-amp is ideal, find v_{out}.



13	Calculate v_{out} [2]

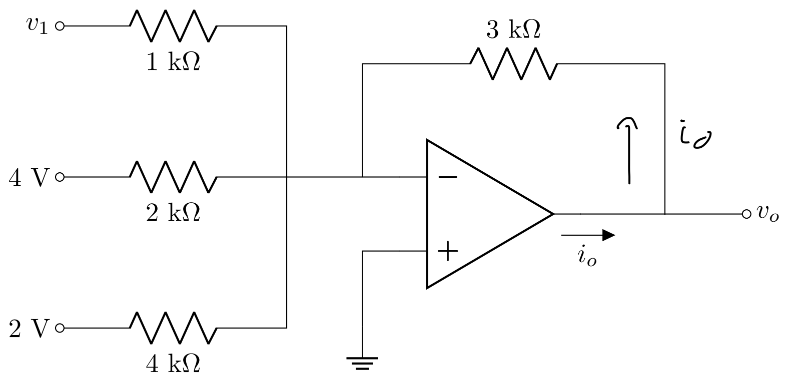
Question 14 [2]
Assuming the op-amp is ideal, calculate v_o.



14	v_o [2]

Questions 15 to 16 [4]

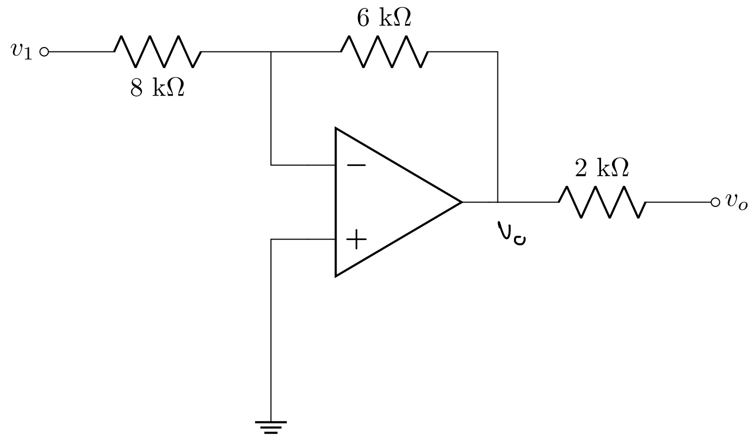
Assume the op-amp is ideal.



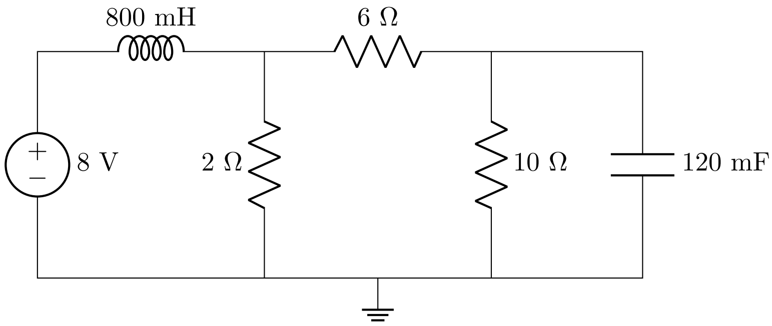
15	If $v_o = 1.5$ V, find v_1 [2]
16	If $v_1 = 3$ V, find i_o [2]

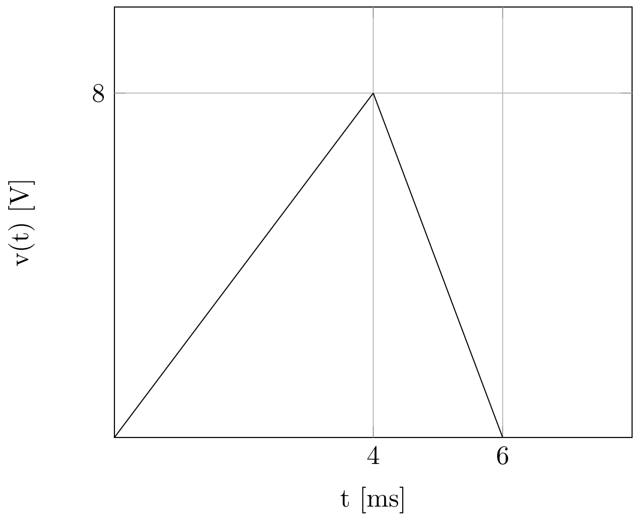
Question 17 [2]

Assume the op-amp is ideal and find the gain v_o/v_1.

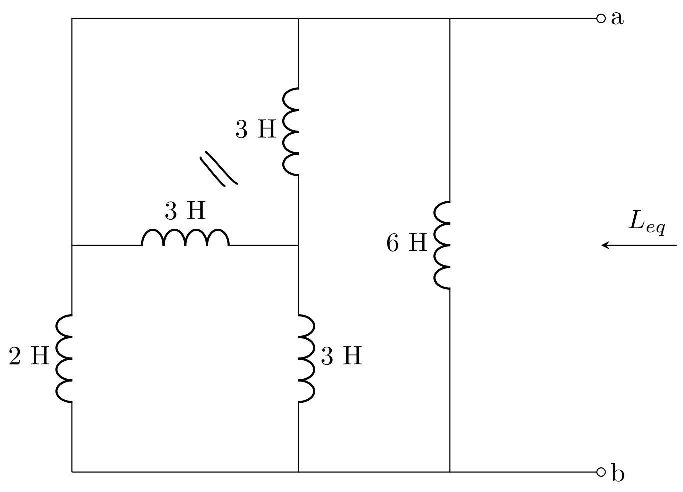


17	v_o/v_1 [2]

Questions 18 to 19 [4]	
Assuming the circuit is under steady-state dc conditions, find the energy stored in the capacitor ω_C, and inductor ω_L, respectively.	
	
18	ω_C [2]
19	ω_L [2]

Question 20 [2]	
A 100 μF capacitor has a voltage $v(t)$ as shown in the graph below. What is the current flowing through it at t = 5 ms $i(5\text{ ms})$?	
	
20	$i(5\text{ ms})$ [2]

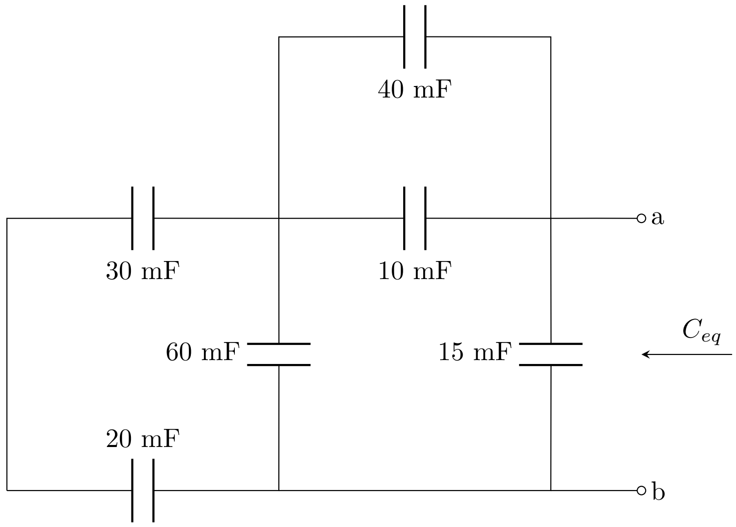
Question 21 [2]
Find the equivalent inductance L_{eq} between terminals a - b.



21	L_{eq} [2]

Question 22 [2]

Find the equivalent capacitance C_{eq} between terminals a - b.



22	C_{eq} [2]

Questions 23 to 24 [2]	
Simplify the following sinusoid so that it is in the form, $\mathbf{I} = A\cos(\omega t + \theta)$. $$\mathbf{I} = 2\sin(40t - 30^\circ) - 4\cos(40t + 110^\circ)$$ Find the period T and angle θ in degrees.	
23	T [1]
24	θ [1]

Question 25 [2]
Simplify **V**. $$\mathbf{V} = \frac{(4\angle -35^\circ)(3 - j5)^*}{4 + j6}$$

25	**V** [2]

Question 26 [2]

Determine if $\mathbf{I}_1$ leads or lags $\mathbf{I}_2$ and by how much?

$$\mathbf{I}_1 = 2\sin(18t - 80^\circ)$$

$$\mathbf{I}_2 = -3\cos(18t + 150^\circ)$$

26	Write only the corresponding number (1, 2, 3 or 4) from the multiple choice. [2]

1. $\mathbf{I}_1$ leads $\mathbf{I}_2$ by 40°
2. $\mathbf{I}_1$ lags $\mathbf{I}_2$ by 40°
3. $\mathbf{I}_1$ leads $\mathbf{I}_2$ by 140°
4. $\mathbf{I}_1$ lags $\mathbf{I}_2$ by 140°

Total Section 1: [46]

Grand Total : [46]

PRACTICE ANSWER SHEET

Assessment ID	2	0	2	4	E	B	N	1	1	1	S	0	2

			Answer	Unit
SECTION 1				
01	v_{1A}	=	10	
02	v_{1B}	=		
03	i_2	=		
04	B	=		No unit required
05	R_1	=		
06	V_{Th}	=		
07	R_{Th}	=		
08	P_{max}	=		
09	I_N	=		
10	R_N	=		
11	v_o	=		
12	R_1	=		
13	v_{out}	=		
14	v_o	=		
15	v_1	=		
16	i_o	=		
17	v_o/v_1	=		No unit required
18	ω_C	=		
19	ω_L	=		
20	$i(5\text{ ms})$	=		
21	L_{eq}	=		
22	C_{eq}	=		
23	T	=		
24	θ	=		No unit required
25	V	=		No unit required
26	Multiple choice:			No unit required